Karlsruhe University of Applied Sciences

Department of Computer Science and Business Information Systems

Business Information Systems

BACHELOR´S THESIS

Efficiency vs. Performance in Green AI: An Empirical Investigation of Energy Consumption, Carbon Footprint, and Predictive Accuracy of Classification Algorithms at Varying Dataset Sizes.

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Chapter 1

Introduction

1.1 Problem statement and societal relevance

The current trend in machine learning is moving towards increasingly large models and datasets [1]. Although deep learning approaches often achieve the highest predictive accuracy, their energy demand and associated CO_2 emissions increase disproportionately [2, 3]. In an era marked by strict sustainability guidelines and rising energy costs, a critical question arises: At what point is a computationally "simple" model ecologically more viable than a highly complex one?

The primary objective of this thesis is to determine whether and when it is worthwhile to deploy more complex models, considering not only their predictive accuracy but also their associated energy consumption. To achieve this, a Logistic Regression, a Random Forest, an XGBoost model and a Multilayer Perceptron are compared under equal conditions across three datasets of varying sizes. This comparative analysis aims to provide a clearer understanding of the thresholds at which simpler models represent the better choice. Additionally, the study investigates how modifying a Multilayer Perceptron architecture, specifically varying the number of neurons and hidden layers, directly impacts energy consumption and CO_2 emissions.

To address the problem statement and achieve the outlined objectives, this thesis investigates the overarching question: *How does the relationship between energy consumption and predictive accuracy change for Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron when model complexity and dataset size are systematically varied?*

Specifically, the following sub-questions are examined:

- How do energy consumption (in kWh) and CO_2 emissions change when scaling a deep learning architecture (variation of neuron count)?
- Under which conditions (dataset size, model complexity) does the resource consumption of complex models exceed the achievable accuracy gain compared to simpler algorithms?
- How does the energy consumption differ between CPU-based training (Random Forest) and GPU-accelerated training (XGBoost, MLP) for the same classification task?

1.2 Motivation: When does a simpler model pay off?

1.3 Research questions and objectives

1.4 Structure of the Thesis

The remainder of this thesis is structured as follows: Chapter 2 provides the theoretical background, distinguishing between Green AI and Red AI, and introducing the relevant sustainability and classification metrics. Chapter 3 details the methodology, including dataset selection and the experimental hardware setup. Chapter 4 outlines the implementation process, focusing on data preprocessing, model training, and the logging of resource consumption. The empirical results are analyzed in Chapter 5, evaluating both performance and resource utilization across scaling dataset sizes. Chapter 6 discusses the findings, highlighting the economic and environmental implications. Finally, Chapter 7 concludes the thesis and provides actionable recommendations for sustainable machine learning practices.

$$S = \frac{F_1}{1 + \lambda_1 \cdot t_{\text{scaled}} + \lambda_2 \cdot c_{\text{scaled}}}$$

Chapter 2

Theoretical Background

Chapter 3

Related Work

We also gathered several other characteristics of each data set, including number of categorical features (NCAT), number of continuous features (NCONT), proportion of features that are categorical (RCAT), number of target classes (NC), minimum single-class proportion (CMIN), and class imbalance (IC). The latter is a measure of the difference in actual and expected proportion of the least common target class (L) and its expected value (LE); i.e., I = (LE – L)/LE. These values are shown in Table 2.

Table 2: Additional Data Set Characteristics

Set	N_{CAT}	N_{CONT}	R_{CAT}	N_C	C_{MIN}	I_C
ANN	32	6	0.842	5	0.001	0.956
BCW	0	9	0.000	2	0.350	0.300
BMK	10	10	0.500	2	0.113	0.775
CAR	0	21	0.000	3	0.083	0.752
CVE	6	0	1.000	4	0.038	0.850
CVR	16	0	1.000	2	0.386	0.228
DCC	3	20	0.130	2	0.221	0.558
DRB	0	19	0.000	2	0.469	0.062
DRE	0	64	0.000	10	0.098	0.016
ILP	1	9	0.100	2	0.286	0.427
ION	0	34	0.000	2	0.359	0.282
LRE	0	16	0.000	26	0.037	0.046
MAG	0	10	0.000	2	0.322	0.357
MPE	3	77	0.038	8	0.092	0.259
MSH	22	0	1.000	2	0.482	0.036
OCD	0	5	0.000	2	0.231	0.538
PHW	30	0	1.000	2	0.443	0.114
QBD	0	41	0.000	2	0.337	0.325
SSG	0	3	0.000	2	0.208	0.585
WFR	0	24	0.000	4	0.060	0.760

During the MLP training experiments, several configurations were evaluated to optimize performance. Initially, the model was trained using only the CPU, which proved too slow for practical use. Switching to CUDA-based GPU training introduced a new issue: the default batch size of 128 provided by `scikit-learn` resulted in training times even slower than on CPU, due to the overhead of frequent small data transfers to the GPU.

Increasing the batch size to 8,192 significantly reduced training time. Further tuning of this parameter did not yield improvements, so this value was retained. Additionally, `pin_memory=True` was evaluated, which locks data in RAM to accelerate CPU-to-GPU transfers. However, this option also led to slower training, likely because the MLP architecture is not complex enough to benefit from the reduced transfer overhead. These experiments led to the final configuration used for all reported MLP results.

Chapter 4

Experimental Design

This chapter outlines the experimental design used to evaluate the ecological efficiency of different machine learning classification algorithms. It describes the datasets, models, evaluation metrics, hyperparameter tuning and measurement setup that form the basis of this empirical analysis.

4.1 Research Questions

The main research question that this study aims to answer is: *How does the trade-off between energy consumption and predictive performance vary across Random Forest, XGBoost and MLP models when the complexity of the models and the size of the dataset are systematically varied?* The following sub-questions will also be addressed in the course of this thesis:

- Under what conditions do the resource consumption of more complex models exceed the achievable accuracy gain over simpler baselines?
- How does energy consumption and CO2 output scale with increasing neural network complexity (number of layers and neurons)?
- How does energy consumption differ between CPU-based and GPU-based training for equivalent tasks?
- How does reducing the number of training instances on large datasets affect model accuracy and energy consumption across different algorithms?

To answer these questions multiple classification algorithms and datasets were used in the experimental phase of this empirical analysis.

4.2 Datasets

The following three datasets, ranging in size from small to large, were selected to assess the effect of dataset size on the relationship between model accuracy, energy consumption and carbon footprint.

The *Wine* dataset contains 178 instances across 13 features, where each feature describes a chemical property of wines from the same region in Italy, derived from three different

Table 4.1: Overview of datasets used in this analysis

Dataset	**Instances**	**Features**	**Classes**	**Scale**
Wine [4]	178	13	3	Small
Default of Credit Card Clients [5]	30,000	23	2	Medium
HIGGS [6]	11,000,000	28	2	Large

cultivars.

The *Default of Credit Card Clients* (hereafter: *Credit*) dataset contains 30,000 instances across 23 features and focuses on predicting payment default among credit card clients in Taiwan.

The *HIGGS* dataset is the largest dataset used in this study, comprising 11,000,000 instances across 28 features, seven of which are high-level features derived by physicists to help discriminate between the two classes [7]. The classification goal is to distinguish between signal events produced by Higgs bosons and background noise processes.

Collectively, these three datasets span several orders of magnitude in size, thus enabling a systematic evaluation of how model performance and energy consumption scale across varying levels of data complexity.

4.3 Models

Five classification models were selected to represent a broad spectrum of algorithmic complexity, ranging from linear baselines to tree-based ensemble methods and neural networks.

Table 4.2: Overview of classification models used in this study

Model	**Type**	**Device**	**Role**
Logistic Regression [8]	Linear	CPU	Baseline
Random Forest [9]	Ensemble	CPU	Mid-complexity
XGBoost [10]	Gradient Boosting	CPU & GPU	Mid-complexity
Multilayer Perceptron	Neural Network	GPU	High-complexity

Logistic regression is a linear classification algorithm that uses the *sigmoid function* to model the probability of class membership, mapping any real-valued input to a value between zero and one. Despite its simplicity, logistic regression often serves as a competitive baseline, making it well suited to quantifying the accuracy gains achieved by more complex models relative to their additional resource consumption.

By contrast, Random Forest is an ensemble method that constructs a large number of decision trees during training, with each tree built on a random subset of the data and features. The final prediction is determined by aggregating the outputs of all the individual trees in a process known as *bagging*. Random Forest does not support GPU-accelerated training and is therefore run exclusively on the CPU.

XGBoost is a gradient boosting algorithm that, like Random Forest, builds an ensemble of decision trees. Unlike *bagging*, however, XGBoost constructs trees sequentially, where each tree attempts to correct the errors of its predecessor. This sequential approach, combined with regularization techniques, often results in higher predictive accuracy than Random Forest. In this study, XGBoost was evaluated in two configurations: CPU-based and GPU-accelerated training, allowing for a direct comparison of energy consumption and training time between the two hardware setups.

The Multilayer Perceptron (hereafter: MLP) was selected as the most complex model in this study, representing the deep learning category. An MLP is a feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer, where each layer applies a weighted sum of its inputs followed by a non-linear activation function, enabling the network to learn complex, non-linear relationships in the data. Training is performed via backpropagation, iteratively updating network weights based on the gradient of the loss function. Its highly configurable architecture, particularly the number of hidden layers and neurons per layer, makes it well suited for the architectural complexity experiments conducted in this study, where these parameters are systematically varied to assess their impact on predictive performance and energy consumption.

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